Modelling a 3D Printer

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0 Introduction

3D printers have the basic task of extruding filament at a given x, y, z coordinate at time t. The mechanism that achieves that movement determines the machine's costs, print quality, and robustness. Out of many different options, the Core XY mechanism is a popular choice for modern printers. Unlike Cartesian designs, where one motor drives one axis, here each motor simultaneously changes the x and y position of the tool head (see Figure 1). This results in a more complex kinematic model, but allows a sturdier casing and faster printing speeds. Our task is to model and simulate the system.

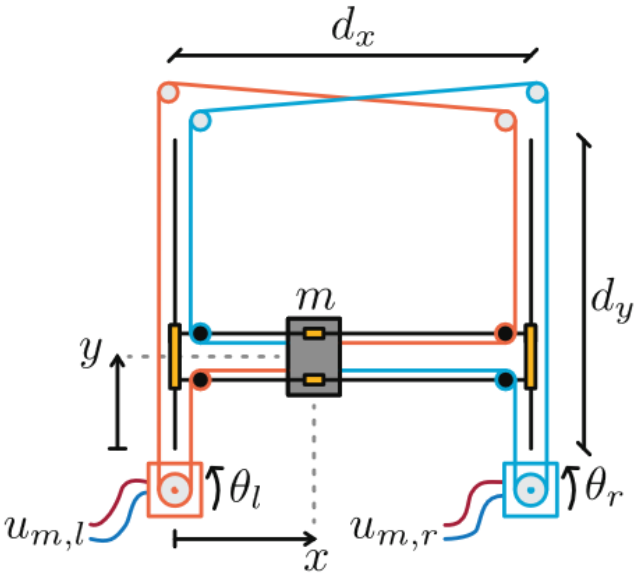


Figure 1: Schematic of the Core XY mechanism.

0.1 Symbol Table

Table 1: Symbols and parameter values.

Description	Symbol	Value
Motor angle left	θ_L	–
Motor voltage left	$u_{m,L}$	–
Motor current left	i_L	–
Motor radius	r_m	11 mm
Motor inductance	L_m	10 mH
Motor resistance	R_m	90 Ω
Motor inertia	J_m	8×10^{-6} kg m^2
Motor constant	K_m	0.07 N m/A
Tool head position	x, y	–
Tool head mass	m	60 g
Nozzle voltage	u_n	–
Nozzle inductor current	$i_{L,n}$	–
Nozzle capacitance	C_n	–
Nozzle inductance	L_n	–
Capacitor voltage	u_c	–
Nozzle thermal capacity	$C_{t,n}$	–
Nozzle thermal resistance	$R_{t,n}$	9 K/W
Nozzle resistance	R_n	8 Ω
Resistor thermal capacity	C_r	–
Resistor temperature	T_r	–
Thermal resistance to environment	R_{env}	–
Reference resistance	R_0	–
Reference temperature	T_0	–
Temperature coefficient of resistance	α	–
Nozzle temperature	T_n	240 °C
Environmental temperature	T_e	26.85 °C
Bearing friction rolling	μ_r	2.7×10^{-8} N m s/rad
Bearing friction linear x-direction	$\mu_{l,x}$	0.002 N s/m
Bearing friction linear y-direction	$\mu_{l,y}$	0.001 N s/m
Bearing radius	r_b	3 mm
Print bed temperature	T_b	105 °C
Print area dimension x	d_x	29 cm
Print area dimension y	d_y	29 cm

1 Physical Modelling

1.1 Kinematics

The first task is to derive the direct kinematics of the Core XY mechanism. This is slightly complicated by the fact that rotating one motor will move the tool head in both x and y directions. By removing the right motor, we can see that by moving the tool head in the positive x direction **or** in the positive y direction, the left motor will rotate in the positive direction. We can draw similar conclusions for the right motor, for which we get the following equations:

$$\begin{aligned} \Delta L &= \Delta x + \Delta y \\ \Delta R &= \Delta x - \Delta y \\ \Delta x &= \frac{\Delta L + \Delta R}{2} \\ \Delta y &= \frac{\Delta L - \Delta R}{2} \end{aligned} \tag{1.1}$$

With ΔL and ΔR being the length of belt moved by the motors, and Δx and Δy being the change in position of the tool head.

$$\begin{aligned} \Delta L &= r_m \, \theta_L \\ \Delta R &= r_m \, \theta_R \end{aligned} \tag{1.2}$$

If we assume that in the beginning the motors are at $\theta_L = \theta_R = 0$ and the tool head is at $(0, 0)$, we get the following equations:

$$\begin{aligned} x(\theta_L, \theta_R) &= \frac{r_m \theta_L + r_m \theta_R}{2} = \frac{r_m}{2} \left(\theta_L + \theta_R\right) \\ y(\theta_L, \theta_R) &= \frac{r_m \theta_L - r_m \theta_R}{2} = \frac{r_m}{2} \left(\theta_L - \theta_R\right) \end{aligned} \tag{1.3}$$

$$\begin{aligned} \dot{x}(\dot{\theta}_L, \dot{\theta}_R) &= \frac{r_m}{2} \left(\dot{\theta}_L + \dot{\theta}_R\right) \\ \dot{y}(\dot{\theta}_L, \dot{\theta}_R) &= \frac{r_m}{2} \left(\dot{\theta}_L - \dot{\theta}_R\right) \end{aligned} \tag{1.4}$$

1.2 Dynamics

In the next part we model the dynamics of the 3D printer system by applying the Euler–Lagrange equation. Additionally, we will split the problem into a mechanical and an electrical one, with the angular velocity of the motor shafts connecting them.

Because of that, we choose generalised coordinates:

$$q = \begin{bmatrix} x \\ y \\ q_L \\ q_R \end{bmatrix} \qquad \dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix} \tag{1.5}$$

We can also then split them among our two problems:

$$\dot{q}_m = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} \text{ (mechanical part)}, \qquad \dot{q}_e = \begin{bmatrix} i_L \\ i_R \end{bmatrix} \text{ (electrical part)} \tag{1.6}$$

1.2.1 Mechanical Part

Kinetic Energy

Firstly, we derive the kinetic energy terms. In the *mechanical* part, the only mass is the tool head. With that we have:

$$K_{\text{mech}} = \tfrac{1}{2} m(\dot{x}^2 + \dot{y}^2) \tag{1.7}$$

As the whole system is planar, we can see that there are no potential energy terms as it will be constant.

Dissipative Energy

Secondly, we find the dissipative energy terms. Each motor has 4 rotational bearings, and 4 linear bearings for the whole system, two along each axis. We define r_b as the bearing radius, later it will be kept the same as the motor radius for simplicity. From (1.3), we can get transformations (1.8) with trivial time derivatives.

This needs to be corrected, not all bearings roll at all times, they depend on what motor rotates

$$\theta_{b,L} = \frac{x+y}{r_b}, \qquad \theta_{b,R} = \frac{x-y}{r_b} \tag{1.8}$$

Then, for each side we get rotational dissipative energy:

$$D_L = 4 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,L}^2 \, \mu_r = 2\mu_r \frac{1}{r_b^2} \left[\dot{x}^2 + 2\dot{x}\dot{y} + \dot{y}^2\right]$$
$$D_R = 4 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,R}^2 \, \mu_r = 2\mu_r \frac{1}{r_b^2} \left[\dot{x}^2 - 2\dot{x}\dot{y} + \dot{y}^2\right]$$

Then simply combining them:

$$\boxed{D_{\text{rot}} = 4 \, \frac{\mu_r}{r_b^2} \left[\dot{x}^2 + \dot{y}^2\right]}$$

Similarly, we get the linear dissipative energy:

$$D_{\text{lin}} = 2 \cdot \tfrac{1}{2} \mu_{l,x} \, \dot{x}^2 + 2 \cdot \tfrac{1}{2} \mu_{l,y} \, \dot{y}^2 = \mu_{l,x} \, \dot{x}^2 + \mu_{l,y} \, \dot{y}^2$$

Then, combining the terms we get:

$$D_{\text{mech}} = \left[4 \, \frac{\mu_r}{r_b^2} + \mu_{l,x}\right] \dot{x}^2 + \left[4 \, \frac{\mu_r}{r_b^2} + \mu_{l,y}\right] \dot{y}^2 \tag{1.9}$$

1.2.2 Electrical Part

Now we model the energies of the electrical part of the system. Keep in mind that both motors are the same. Below we can see the equivalent circuit of a DC motor:

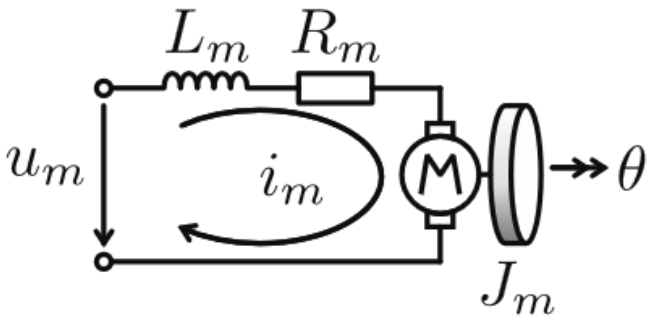


Figure 2: Equivalent circuit of a DC motor.

Kinetic Energy

The inductor and the inertial element will contribute to the kinetic energy of this part of the system. Similarly to (1.8), using the motor shaft radius:

$$K_L = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \dot{\theta}_L^2 = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y}\right)$$
$$K_R = \tfrac{1}{2} i_R^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 - 2\dot{x}\dot{y}\right)$$

Combining them we get:

$$K_{\text{el}} = \tfrac{1}{2} L_m \left[i_R^2 + i_L^2\right] + J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2\right) \tag{1.10}$$

As there are no capacitors in the system, there is no potential energy change as in the mechanical part.

Dissipative Energy

The only dissipative term in the electrical part of the system is the resistor:

$$\begin{gathered} D_{\text{el},L} = \tfrac{1}{2} i_L^2 R_m, \qquad D_{\text{el},R} = \tfrac{1}{2} i_R^2 R_m \\ D_{\text{el}} = \tfrac{1}{2} R_m \left(i_L^2 + i_R^2\right) \end{gathered} \tag{1.11}$$

Matrix Equations

Finally, we realise that the system is linear, and write the matrix expressions:

$$K = \tfrac{1}{2}\, \dot{q}^\top \mathbf{M}\, \dot{q}, \qquad D = \tfrac{1}{2}\, \dot{q}^\top \mathbf{N}\, \dot{q} \tag{1.12}$$

With corresponding matrices:

$$\mathbf{M} = \begin{bmatrix} m + J_m \frac{2}{r_m^2} & 0 & 0 & 0 \\ 0 & m + J_m \frac{2}{r_m^2} & 0 & 0 \\ 0 & 0 & L_m & 0 \\ 0 & 0 & 0 & L_m \end{bmatrix}$$
$$\mathbf{N} = \begin{bmatrix} 8\frac{\mu_r}{r_b^2} + 2\mu_{l,x} & 0 & 0 & 0 \\ 0 & 8\frac{\mu_r}{r_b^2} + 2\mu_{l,y} & 0 & 0 \\ 0 & 0 & R_m & 0 \\ 0 & 0 & 0 & R_m \end{bmatrix} \tag{1.13}$$

1.2.3 Generalised Force

Mechanical Coordinates

The non-conservative forces in the mechanical part are given by the motor torques τ_L and τ_R. The virtual work they perform is:

$$\Delta W = \tau_L \, \Delta\theta_L + \tau_R \, \Delta\theta_R$$

With the familiar transformation, analogous to (1.8) but using the motor radius r_m, we get:

$$\Delta W = \frac{\tau_L + \tau_R}{r_m} \, \Delta x + \frac{\tau_L - \tau_R}{r_m} \, \Delta y$$

Applying the motor law $\tau = K_m i$, the generalised forces for x and y are:

$$Q_x = \frac{K_m}{r_m} \left(i_L + i_R \right), \qquad Q_y = \frac{K_m}{r_m} \left(i_L - i_R \right) \tag{1.14}$$

Electrical Coordinates

The non-conservative forces in the electrical part are the input voltages and the back-EMF of the motors:

$$\Delta W = \left(u_{m,L} - K_m \dot{\theta}_L \right) \Delta q_L + \left(u_{m,R} - K_m \dot{\theta}_R \right) \Delta q_R$$

Using the same transformation, we get the generalised forces for q_L and q_R:

$$Q_{q_L} = u_{m,L} - \frac{K_m}{r_m} \left(\dot{x} + \dot{y} \right), \qquad Q_{q_R} = u_{m,R} - \frac{K_m}{r_m} \left(\dot{x} - \dot{y} \right) \tag{1.15}$$

Lagrange Equations

Using the fact that this is a linear system and there are no potential energies we get:

$$\mathcal{L} = K - V = \tfrac{1}{2} \, \dot{q}^\top \mathbf{M} \, \dot{q} - V, \qquad V = 0 \quad \Rightarrow \quad \mathcal{L} = \tfrac{1}{2} \, \dot{q}^\top \mathbf{M} \, \dot{q}$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} = \frac{d}{dt} \left(\mathbf{M} \dot{q} \right) = \mathbf{M} \ddot{q}$$
$$\frac{\partial \mathcal{L}}{\partial q} = 0$$
$$\frac{\partial D}{\partial \dot{q}} = \mathbf{N} \dot{q}$$

And finally solving the Euler–Lagrange equation:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = -\frac{\partial D}{\partial \dot{q}} + \mathbf{Q}, \qquad \mathbf{Q} = \begin{bmatrix} Q_x \\ Q_y \\ Q_{q_L} \\ Q_{q_R} \end{bmatrix}$$

$$\mathbf{M}\ddot{q} + \mathbf{N}\dot{q} = \mathbf{Q} \tag{1.16}$$

Expanding back out of matrix form we get:

$$\begin{aligned}
\left(m + J_m \frac{2}{r_m^2}\right)\ddot{x} + \left(8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}\right)\dot{x} - \frac{K_m}{r_m} i_L - \frac{K_m}{r_m} i_R &= 0 \\
\left(m + J_m \frac{2}{r_m^2}\right)\ddot{y} + \left(8\frac{\mu_r}{r_b^2} + 2\mu_{l,y}\right)\dot{y} - \frac{K_m}{r_m} i_L + \frac{K_m}{r_m} i_R &= 0 \\
L_m \frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\dot{x} + \frac{K_m}{r_m}\dot{y} - u_{m,L} &= 0 \\
L_m \frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\dot{x} - \frac{K_m}{r_m}\dot{y} - u_{m,R} &= 0
\end{aligned}$$

Or with substitutions, we get the final form of the four differential equations:

$$m_{\text{eq}} = m + \frac{2J_m}{r_m^2}, \qquad b_x = 8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}, \qquad b_y = 8\frac{\mu_r}{r_b^2} + 2\mu_{l,y}$$

$$\boxed{\begin{aligned}
\ddot{x} &= -\frac{b_x}{m_{\text{eq}}}\dot{x} + \frac{K_m}{m_{\text{eq}} r_m}\left(i_L + i_R\right) \\
\ddot{y} &= -\frac{b_y}{m_{\text{eq}}}\dot{y} + \frac{K_m}{m_{\text{eq}} r_m}\left(i_L - i_R\right) \\
\frac{di_L}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x} + \dot{y}\right) - \frac{R_m}{L_m} i_L + \frac{u_{m,L}}{L_m} \\
\frac{di_R}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x} - \dot{y}\right) - \frac{R_m}{L_m} i_R + \frac{u_{m,R}}{L_m}
\end{aligned}} \tag{1.17}$$

Or in **state-space** form:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix}, \qquad \dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \ddot{x} \\ \ddot{y} \\ \dfrac{di_L}{dt} \\ \dfrac{di_R}{dt} \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} u_{m,L} \\ u_{m,R} \end{bmatrix}$$

$$\dot{\mathbf{x}} = \mathbf{A}\,\mathbf{x} + \mathbf{B}\,\mathbf{u} \tag{1.18}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{b_x}{m_{\text{eq}}} & 0 & \frac{K_m}{m_{\text{eq}} r_m} & \frac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & 0 & -\frac{b_y}{m_{\text{eq}}} & \frac{K_m}{m_{\text{eq}} r_m} & -\frac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & -\frac{K_m}{r_m L_m} & -\frac{K_m}{r_m L_m} & -\frac{R_m}{L_m} & 0 \\ 0 & 0 & -\frac{K_m}{r_m L_m} & \frac{K_m}{r_m L_m} & 0 & -\frac{R_m}{L_m} \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{L_m} & 0 \\ 0 & \frac{1}{L_m} \end{bmatrix}$$

1.2.4 Decoupling the x and y Dynamics

To decouple the system, we introduce a set of useful substitutions:

$$\boxed{\begin{aligned} i_x &= i_L + i_R, & u_{m,x} &= u_{m,L} + u_{m,R} \\ i_y &= i_L - i_R, & u_{m,y} &= u_{m,L} - u_{m,R} \\ i_L &= \frac{i_x + i_y}{2}, & u_{m,L} &= \frac{u_{m,x} + u_{m,y}}{2} \\ i_R &= \frac{i_x - i_y}{2}, & u_{m,R} &= \frac{u_{m,x} - u_{m,y}}{2} \end{aligned}} \tag{1.19}$$

By substituting our new decoupled currents we get:

$$\begin{aligned} \frac{di_L}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} + \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} - \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m} i_x - \frac{R_m}{2L_m} i_y + \frac{1}{2L_m}\left(u_{m,x} + u_{m,y}\right) \\ \frac{di_R}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} - \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} + \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m} i_x + \frac{R_m}{2L_m} i_y + \frac{1}{2L_m}\left(u_{m,x} - u_{m,y}\right) \end{aligned}$$

Then by adding and subtracting the two equations to cancel terms we get:

$$\begin{aligned} \frac{di_x}{dt} &= -\frac{2K_m}{r_m L_m}\dot{x} - \frac{R_m}{L_m} i_x + \frac{1}{L_m} u_{m,x} \\ \frac{di_y}{dt} &= -\frac{2K_m}{r_m L_m}\dot{y} - \frac{R_m}{L_m} i_y + \frac{1}{L_m} u_{m,y} \end{aligned} \tag{1.20}$$

And by substituting (1.19) into the first two equations of (1.17):

$$\begin{aligned} \ddot{x} &= -\frac{b_x}{m_{\text{eq}}}\dot{x} + \frac{K_m}{m_{\text{eq}} r_m} i_x \\ \ddot{y} &= -\frac{b_y}{m_{\text{eq}}}\dot{y} + \frac{K_m}{m_{\text{eq}} r_m} i_y \end{aligned} \tag{1.21}$$

And finally:

$$\tilde{\mathbf{x}} = \begin{bmatrix} x \\ \dot{x} \\ i_x \\ y \\ \dot{y} \\ i_y \end{bmatrix}, \qquad \dot{\tilde{\mathbf{x}}} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dfrac{di_x}{dt} \\ \dot{y} \\ \ddot{y} \\ \dfrac{di_y}{dt} \end{bmatrix}, \qquad \tilde{\mathbf{u}} = \begin{bmatrix} u_{m,x} \\ u_{m,y} \end{bmatrix}$$

$$\dot{\tilde{\mathbf{x}}} = \tilde{\mathbf{A}}\,\tilde{\mathbf{x}} + \tilde{\mathbf{B}}\,\tilde{\mathbf{u}} \tag{1.22}$$

$$\tilde{\mathbf{A}} = \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -\dfrac{b_x}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} & 0 & 0 & 0 \\ 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -\dfrac{b_y}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & 0 & 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} \end{array}\right], \qquad \tilde{\mathbf{B}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \dfrac{1}{L_m} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & \dfrac{1}{L_m} \end{bmatrix}$$

1.3 Drawing Shapes

Rearranging the decoupled equations of motion (1.21), we can express the currents needed to follow a prescribed motion of the tool head:

$$\boxed{\begin{aligned} i_x &= \frac{m_{\text{eq}} r_m}{K_m}\,\ddot{x} + \frac{b_x r_m}{K_m}\,\dot{x} \\ i_y &= \frac{m_{\text{eq}} r_m}{K_m}\,\ddot{y} + \frac{b_y r_m}{K_m}\,\dot{y} \end{aligned}} \tag{1.23}$$

The motor equations for each motor are:

$$\begin{aligned} u_{m,L} &= L_m \frac{di_L}{dt} + R_m i_L + K_m \dot{\theta}_L = L_m \frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\,(\dot{x} + \dot{y}) \\ u_{m,R} &= L_m \frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\,(\dot{x} - \dot{y}) \end{aligned}$$

Adding them, and using (1.19):

$$u_{m,x} = u_{m,L} + u_{m,R} = L_m \left[\frac{di_L}{dt} + \frac{di_R}{dt}\right] + R_m\,[i_L + i_R] + \frac{2K_m}{r_m}\,\dot{x}$$

and similarly by subtracting them, we get the decoupled input voltages, consistent with (1.20):

$$\boxed{\begin{aligned} u_{m,x} &= L_m \frac{di_x}{dt} + R_m i_x + \frac{2K_m}{r_m}\dot{x} \\ u_{m,y} &= L_m \frac{di_y}{dt} + R_m i_y + \frac{2K_m}{r_m}\dot{y} \end{aligned}} \tag{1.24}$$

The original motor voltages then follow from (1.19):

$$u_{m,L} = \tfrac{1}{2}\left(u_{m,x} + u_{m,y}\right), \qquad u_{m,R} = \tfrac{1}{2}\left(u_{m,x} - u_{m,y}\right) \tag{1.25}$$

Since (1.24) contains $\frac{di_x}{dt}$, and (1.23) contains $\ddot{x}$, the voltages depend on the third derivative of the position. Therefore, for any C^3 functions $x(t)$ and $y(t)$ describing the position of the tool head, we can derive an open-loop analytical solution for the input voltages, even if those voltages are not realistic.

Ellipse

An ellipse with centre (h, k) and semi-axes a and b has the parametric form:

$$x(t) = h + a\cos t, \qquad y(t) = k + b\sin t \tag{1.26}$$

We also need the derivatives up to the third order:

$$\begin{aligned} \dot{x} &= -a\sin t, \quad \ddot{x} = -a\cos t, \quad \dddot{x} = a\sin t \\ \dot{y} &= b\cos t, \quad \ddot{y} = -b\sin t, \quad \dddot{y} = -b\cos t \end{aligned}$$

Substituting into (1.23):

$$\begin{aligned} i_x(t) &= -\frac{m_{\text{eq}} r_m}{K_m} a\cos t - \frac{b_x r_m}{K_m} a\sin t = -\frac{r_m}{K_m} a\left[m_{\text{eq}}\cos t + b_x \sin t\right] \\ i_y(t) &= \frac{r_m}{K_m} b\left[b_y \cos t - m_{\text{eq}}\sin t\right] \end{aligned}$$

And into (1.24):

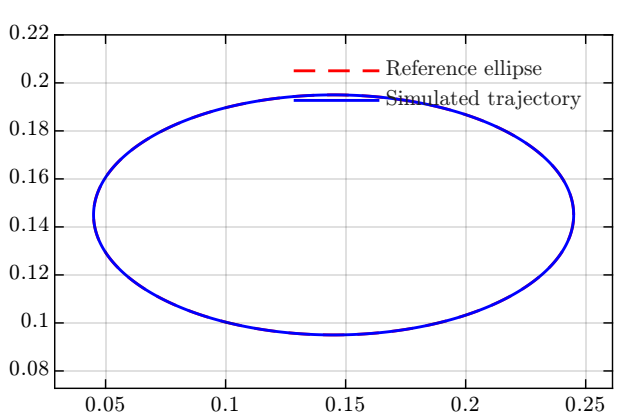
$$\begin{aligned} u_{m,x}(t) &= L_m \frac{r_m}{K_m} a\left[m_{\text{eq}}\sin t - b_x\cos t\right] - R_m \frac{r_m}{K_m} a\left[m_{\text{eq}}\cos t + b_x \sin t\right] - \frac{2K_m}{r_m} a\sin t \\ u_{m,y}(t) &= -L_m \frac{r_m}{K_m} b\left[m_{\text{eq}}\cos t + b_y \sin t\right] + R_m \frac{r_m}{K_m} b\left[b_y\cos t - m_{\text{eq}}\sin t\right] + \frac{2K_m}{r_m} b\cos t \end{aligned}$$

With

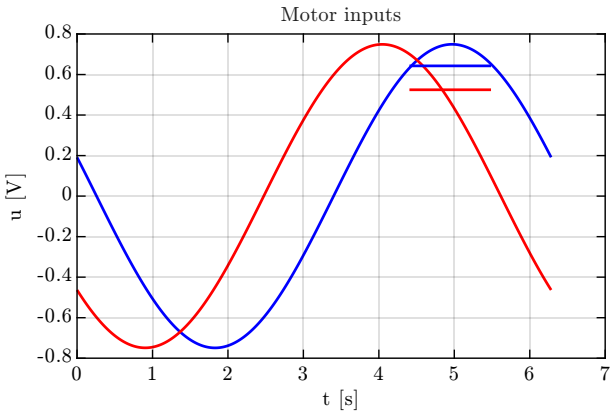
$$u_{m,L}(t) = \tfrac{1}{2}\left(u_{m,x}(t) + u_{m,y}(t)\right), \qquad u_{m,R}(t) = \tfrac{1}{2}\left(u_{m,x}(t) - u_{m,y}(t)\right)$$

we obtain the analytical open-loop solution for drawing the ellipse.

Simulating the given system with parameters defined in Table 1, we get near-perfect alignment with the reference ellipse.



(a) Reference and simulated trajectory.



(b) Open-loop motor voltages.

Figure 3: Open-loop drawing of an ellipse with nominal model parameters.

As this is open-loop control, this good result relies on a good parameter estimation. If we change the parameters of the model by ±5% we can see that an error immediately arises.

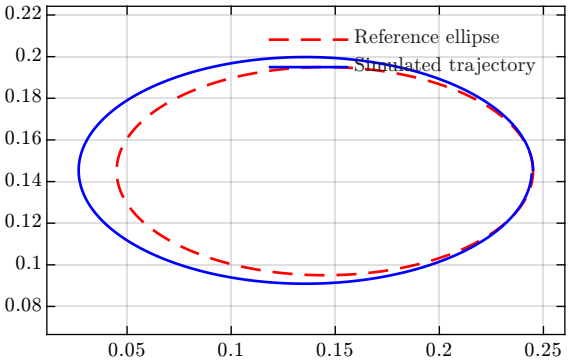
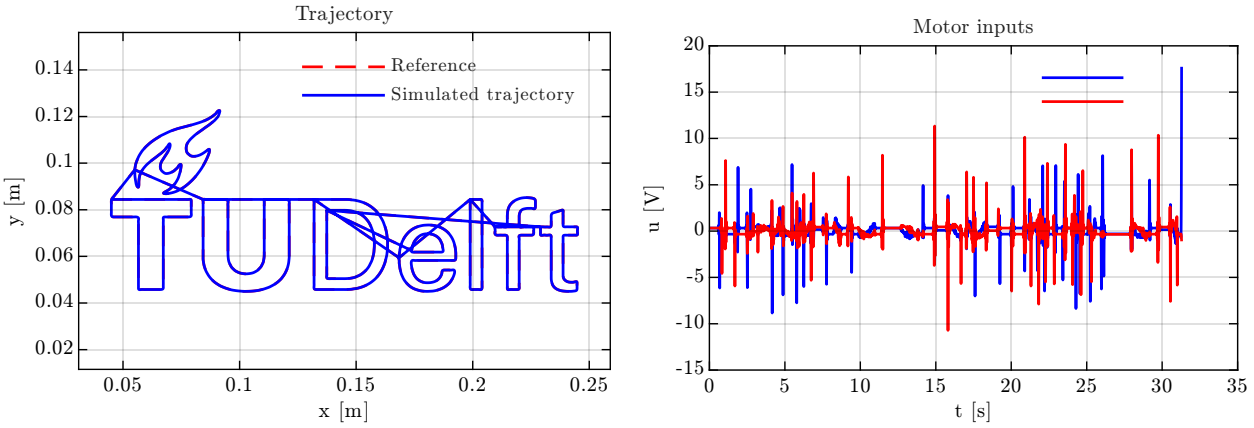


Figure 4: Open-loop drawing of an ellipse with model parameters perturbed by ±5%.

Arbitrary Drawing

As stated in the derivation, any C^3 function used for $x(t)$ and $y(t)$ will allow us to get an analytical solution. With that in mind we can draw arbitrary shapes by approximating them with quartic splines. Using MATLAB's *bwboundaries* function to detect edges of shapes, in combination with *spapi* which gives us an interpolating quartic spline given a set of points, we can draw the *TU Delft* logo[1].

[1]Different letter boundaries are connected by straight lines so the whole drawing can be one input function



(a) Reference and simulated trajectory.

(b) Open-loop motor voltages.

Figure 5: Open-loop drawing of the *TU Delft* logo with nominal model parameters.

Again, like with the ellipse, the logo, because of its sharp corners, is even more sensitive to inaccuracies in the model parameters. If we again perturb the parameters by ±5% we see that control of this form would not be useful anymore.

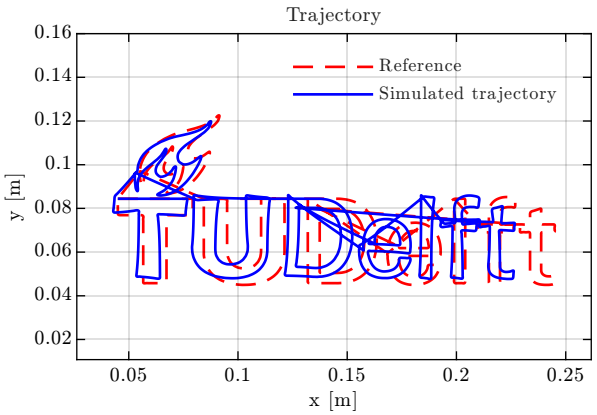


Figure 6: Open-loop drawing of the *TU Delft* logo with model parameters perturbed by ±5%.

1.3.1 Modelling the Nozzle

In the following section we model the nozzle of the 3D printer (Figure 7).

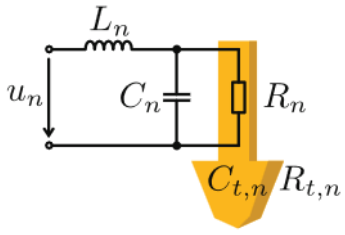


Figure 7: Schematic of the nozzle heating circuit.

add nicer drawing of the system

The system consists of an electrical and thermodynamical part. A circuit consisting of an inductor L_n, capacitor C_n and a heating resistor R_n. We choose the current through the inductor and the voltage across the capacitor, $i_{L,n}$ and u_c respectively, as first two states of our system. By solving the three Kirchhoff equations we get the following two first-order differential equations:

$$\begin{aligned} L_n \frac{di_{L,n}}{dt} &= u_n - u_c \\ C_n \frac{du_c}{dt} &= i_{L,n} - \frac{u_c}{R_n(T_r)} \end{aligned} \tag{1.27}$$

Next, we model the thermal part. Besides the thermal capacitance of the nozzle $C_{t,n}$, given in the task, we also model the thermal capacitance of the resistor C_r[2]. We choose two additional states, temperature of the resistor (T_r) and the temperature of the nozzle (T_n). Losses happening between the resistor and the nozzle are described by $R_{t,n}$, and the heat dissipation to the environment and to the filament is modelled with one variable R_{env}. The connection to the electrical part is the heat dissipated by the resistor $P_r = \frac{u_c^2}{R_n(T_r)}$. This gives us the following two equations which with (1.27) fully describe the system:

$$\begin{aligned} C_{t,n} \frac{dT_n}{dt} &= \frac{T_r - T_n}{R_{t,n}} - \frac{T_n - T_e}{R_{\text{env}}} \\ C_r \frac{dT_r}{dt} &= P_r - \frac{T_r - T_n}{R_{t,n}} \end{aligned} \tag{1.28}$$

with

$$R_n(T_r) = R_0 \left[1 + \alpha \left(T_r - T_0\right)\right] \tag{1.29}$$

where R_0 is the resistance at the reference temperature T_0 and α is the temperature coefficient of resistance.

Due to the square voltage term and the temperature dependence of the resistance the system is very non-linear, although the very different time constants of two systems (thermal states change much slower compared to the electrical) can be exploited. Alternatively, the model can reasonably be simplified by assuming a constant resistance. With that formulation we would get two linear systems connected by a single non-linear term.

[2] If we do not model the thermal capacitance of the resistor, we end up with a differential-algebraic equation (DAE).

1.3.2 Printing a circle of filament

Moving-Boundary Heat Problem

better notation, proper plots, explanation text

Cross-section $a \times a$, element length ΔL; three faces to the environment (T_e), one face to the heating plate (T_p).

Energy balance

$$\partial_t u = -\nabla \cdot \phi + f, \qquad u = \rho c\, T, \qquad \phi = -\kappa \nabla T \qquad \Longrightarrow \qquad \partial_t T = \frac{\kappa}{\rho c}\, \nabla^2 T + \frac{f}{\rho c}$$

$$[u] = \mathrm{J/m^3}, \qquad [\phi] = \mathrm{W/m^2}, \qquad [f] = \mathrm{W/m^3}$$

Heat exchange

$$\Delta A_e = 3a\, \Delta L, \qquad \Delta A_p = a\, \Delta L, \qquad V = a^2\, \Delta L$$

$$\dot{Q} = K_e\, \Delta A_e\, (T_e - T) + K_p\, \Delta A_p\, (T_p - T) \qquad \left[\mathrm{W/(m^2 K)} \cdot \mathrm{m^2} \cdot \mathrm{K} = \mathrm{W}\right]$$

$$f = \frac{\dot{Q}}{V} = \frac{1}{a}\Big[\, 3K_e\, (T_e - T) + K_p\, (T_p - T)\,\Big]$$

Result

$$\boxed{\partial_t T = k\, \partial_{xx} T + \alpha\, T + \beta}$$

$$k = \frac{\kappa}{\rho c}, \qquad \alpha = -\frac{3K_e + K_p}{a\, \rho c}, \qquad \beta = \frac{3K_e T_e + K_p T_p}{a\, \rho c}$$

2. Front Fixing

John Crank, Free and Moving Boundary Problems

$$\xi = \frac{x}{s(t)}, \qquad 0 \le \xi \le 1, \qquad s(t) = s_0 + s_1 t, \qquad \dot{s} = s_1$$

Derivatives

$$\frac{\partial T}{\partial x} = \frac{1}{s}\, \frac{\partial T}{\partial \xi}, \qquad \frac{\partial^2 T}{\partial x^2} = \frac{1}{s^2}\, \frac{\partial^2 T}{\partial \xi^2}$$

$$\left(\frac{\partial T}{\partial t}\right)_x = \frac{\partial T}{\partial \xi}\left(\frac{\partial \xi}{\partial t}\right)_x + \left(\frac{\partial T}{\partial t}\right)_\xi = -\frac{x\, \dot{s}}{s^2}\, \frac{\partial T}{\partial \xi} + \left(\frac{\partial T}{\partial t}\right)_\xi = -\frac{\xi\, \dot{s}}{s}\, \frac{\partial T}{\partial \xi} + \left(\frac{\partial T}{\partial t}\right)_\xi$$

Transformed equation

$$\boxed{\left(\frac{\partial T}{\partial t}\right)_\xi = \frac{\xi\, \dot{s}}{s}\, \frac{\partial T}{\partial \xi} + \frac{k}{s^2}\, \frac{\partial^2 T}{\partial \xi^2} + \alpha\, T + \beta}$$

3. Boundary and Initial Conditions

$$T(1, t) = T_n \quad \text{(nozzle temperature at } x = s(t)\text{)}$$

$$\partial_x T(0, t) = 0 \quad \Longrightarrow \quad \frac{T_1^j - T_0^j}{\Delta \xi} = 0 \quad \Longrightarrow \quad T_0^j = T_1^j$$

$$T_i^0 = T_n, \qquad s(0) = s_0 \text{ (small)}$$

4. Discretisation (implicit)

$$\xi_i = i\,\Delta\xi, \quad \Delta\xi = \tfrac{1}{N}, \quad i = 0,\dots,N; \qquad t_j = j\,\Delta t; \qquad T_i^j \approx T(\xi_i, t_j)$$

$$\frac{\partial T}{\partial t} \approx \frac{T_i^j - T_i^{j-1}}{\Delta t}, \qquad \frac{\partial T}{\partial \xi} \approx \frac{T_{i+1}^j - T_i^j}{\Delta\xi}, \qquad \frac{\partial^2 T}{\partial \xi^2} \approx \frac{T_{i+1}^j - 2T_i^j + T_{i-1}^j}{\Delta\xi^2}$$

$$\gamma_i = \frac{\xi_i\,\dot{s}\,\Delta t}{s\,\Delta\xi}, \qquad g = \frac{k\,\Delta t}{s^2\,\Delta\xi^2}, \qquad s,\ \dot{s} \text{ evaluated at } t_j$$

$$\boxed{T_i^j\left[1 + \gamma_i + 2g - \alpha\,\Delta t\right] = g\,T_{i-1}^j + (\gamma_i + g)\,T_{i+1}^j + T_i^{j-1} + \beta\,\Delta t}$$

Stability

von Neumann probably

5. Linear System at $t = t_j$

$$d_i = 1 + \gamma_i + 2g - \alpha\,\Delta t, \qquad e_i = \gamma_i + g$$

$$\begin{bmatrix} 1 & -1 & & & & \\ -g & d_1 & -e_1 & & & \\ & -g & d_2 & -e_2 & & \\ & & \ddots & \ddots & \ddots & \\ & & & -g & d_{N-1} & -e_{N-1} \\ & & & & 0 & 1 \end{bmatrix} \begin{bmatrix} T_0^j \\ T_1^j \\ T_2^j \\ \vdots \\ T_{N-1}^j \\ T_N^j \end{bmatrix} = \begin{bmatrix} 0 \\ T_1^{j-1} + \beta\,\Delta t \\ T_2^{j-1} + \beta\,\Delta t \\ \vdots \\ T_{N-1}^{j-1} + \beta\,\Delta t \\ T_n \end{bmatrix}$$

6. After the Boundary Stops (Closed Ring)

$$\dot{s} = 0 \ \Rightarrow\ \gamma_i = 0, \qquad s = L, \qquad T_N^j \equiv T_0^j \text{ (first and last node coincide)}$$

$$d = 1 + 2g - \alpha\,\Delta t$$

$$\begin{bmatrix} d & -g & & & -g \\ -g & d & -g & & \\ & \ddots & \ddots & \ddots & \\ & & -g & d & -g \\ -g & & & -g & d \end{bmatrix} \begin{bmatrix} T_0^j \\ T_1^j \\ \vdots \\ T_{N-2}^j \\ T_{N-1}^j \end{bmatrix} = \begin{bmatrix} T_0^{j-1} + \beta\,\Delta t \\ T_1^{j-1} + \beta\,\Delta t \\ \vdots \\ T_{N-2}^{j-1} + \beta\,\Delta t \\ T_{N-1}^{j-1} + \beta\,\Delta t \end{bmatrix}$$

First step after connection:

$$T_0^{j-1} \leftarrow \tfrac{1}{2}\left(T_0^{j-1} + T_N^{j-1}\right)$$

Choose and display the chosen model parameters

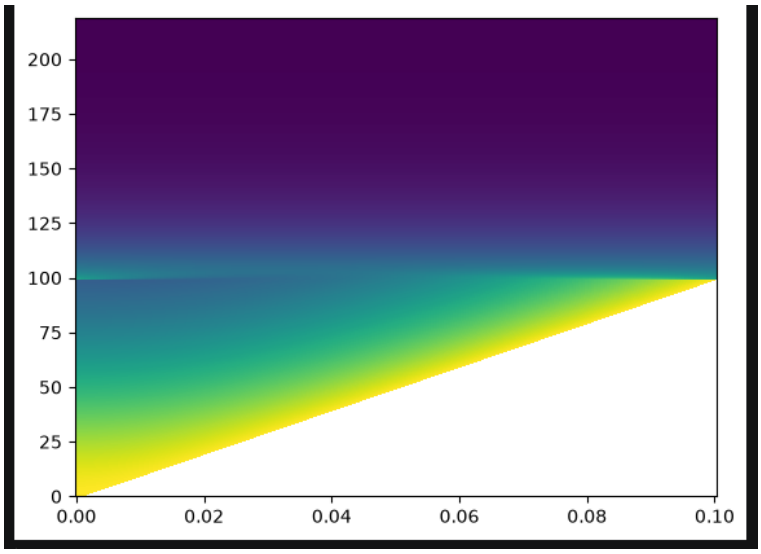


Figure 8: INITIAL PLOT, CHANGE.

2 Finite State Machine

2.1 Filament Replacement

This section presents the high-level control of the filament replacement, modelled as a Mealy machine since the next state and the output (status message and retry counter) both depend on the current state and input. The machine listens to the external events C (a new colour is requested) and F (the current spool is finished) and to the success or failure feedback of the low-level subsystems. It consists of two coupled sub-machines that are updated together: a *printing* machine (idle, running or paused) and a *filament replacement* machine that sequences the steps of Q2.1. A failed step is retried up to three times before the replacement is aborted.

2.1.1 Formal Machine Definition

The FSM is defined as the Mealy machine

$$\mathcal{M} = (Q, \Sigma, \Lambda, \delta, \lambda, q_0). \tag{2.1}$$

States. The state $q = (p, f, r)$ of printing state, filament replacement state and retries left,

$$\begin{aligned} Q &= P \times \mathcal{F} \times R, \\ P &= \{\texttt{IDLE}, \texttt{IN_PROG}, \texttt{PAUSED}\}, \\ \mathcal{F} &= \{\texttt{IDLE}, \texttt{MOVE}, \texttt{WIPE}, \texttt{COOL}, \texttt{LOAD}, \texttt{HEAT}\}, \\ R &= \{0, 1, 2, 3\}, \end{aligned} \tag{2.2}$$

where `MOVE` is moving to the wipe location, `WIPE` is wiping the excess filament, `COOL` is the nozzle heat switched off, `LOAD` is loading the new filament and `HEAT` is the nozzle heat switched on. Saving the head location (step 1 of Q2.1) is the action on entering `MOVE`, and resuming the print (step 7) is the action on leaving `HEAT`. The initial state is $q_0 = (\texttt{IDLE}, \texttt{IDLE}, 3)$.

Inputs. At each step the FSM can accept a pair of inputs $\sigma = (e, y)$, where e represents external event i.e C, F or nothing. y represents the feedback of the outcome of the current filament state being executed. It is represented by ok_k if step k succeeded and fail_k if it failed, for each of the five replacement steps $k \in K = \mathcal{F} \setminus \{\texttt{IDLE}\} = \{\texttt{MOVE}, \texttt{WIPE}, \texttt{COOL}, \texttt{LOAD}, \texttt{HEAT}\}$. For example, $\text{fail}_{\texttt{WIPE}}$ means that the wiping did not succeed.

In total there are 3 external events and $1 + 2 \cdot 5 = 11$ feedback values, so the input alphabet has $3 \cdot 11 = 33$ elements:

$$\begin{aligned} E &= \{C,\ F,\ \varepsilon\}, \\ Y &= \{\varepsilon,\ \text{ok}_{\texttt{MOVE}},\ \text{fail}_{\texttt{MOVE}},\ \text{ok}_{\texttt{WIPE}},\ \text{fail}_{\texttt{WIPE}},\ \ldots,\ \text{ok}_{\texttt{HEAT}},\ \text{fail}_{\texttt{HEAT}}\}, \\ \Sigma &= E \times Y. \end{aligned} \tag{2.3}$$

Outputs. The output alphabet λ represents the set of output messages sent to the user on each transition. For example, “filament change in progress”, “k successful”, “k failure, trying again”, “filament change process failed, initiate filament change again” and “filament change already in progress”.

Transition and output functions. The transition function $\delta : Q \times \Sigma \to Q$ and output function $\lambda : Q \times \Sigma \to \Lambda$ are given by the transition table and below. Here $\text{next}(k)$ is the step after k in the order `MOVE` → `WIPE` → `COOL` → `LOAD` → `HEAT`, and r is the number of retries left, reset to 3 on every new step.

The following cases can occur, depending on the filament replacement state f and the input $\sigma = (e, y)$:

- **Start.** If $f =$ `IDLE` and the event e is C or F, the replacement starts: the machine moves to `MOVE`, and a running print ($p =$ `IN_PROG`) is paused, while an idle or already paused printer keeps its printing state.
- **Event during a replacement.** If $f \neq$ `IDLE` and e is C or F, the event is ignored and the state is unchanged; the output message is "filament change already in progress".
- **Step succeeded.** On ok_k in step $f = k$, the machine moves on to $\text{next}(k)$, with $r = 3$.
- **Last step succeeded.** On $\text{ok}_{\texttt{HEAT}}$ the replacement is finished: the machine returns to $f =$ `IDLE`, and a paused print is resumed ($p =$ `PAUSED` → `IN_PROG`).
- **Step failed, retries left.** On fail_k with $r > 0$, the machine stays in step k and the number of retries is reduced by one, $r' = r - 1$, so the step is tried again.
- **Step failed, no retries left.** On fail_k with $r = 0$, the replacement is aborted: the machine returns to $f =$ `IDLE` with $r = 3$, and the message asks the user to start the filament change again.
- **Anything else.** An input for which the table has no row (e.g. a feedback from a step that is not active) leaves the state unchanged and returns the message `NO MATCHING TRANSITION`.

Each of these cases has its own output message, which is the output λ of the transition.

Implementation. The machine is implemented as a state transition table loaded into a pandas dataframe in `fsm.ipynb`. A snapshot of the table is shown in Table 2.

Every row describes one of the transition cases defined above. The first five columns (printing state, filament state, retries left, external input and system feedback) are the conditions that the current state and input must satisfy, and the last four columns are the output state, the retries left and the output message. The entry `X` represents `DON'T CARE` condition where the current state of this variable does not influence the output in any manner. An empty input cell means that the signal is absent and must match exactly. The rows are checked from top to bottom and the first row whose conditions all hold is applied; if no row matches, the state is left unchanged and the message `NO MATCHING TRANSITION` is returned.

The machine is implemented as a state transition table loaded into a pandas dataframe in `fsm.ipynb`. A snapshot of the table is shown in Table 2.

Table 2: Snapshot of the state transition table (state names abbreviated, `X` is a `DON'T CARE`, empty cells mean no signal).

p	f	e	y	r	p'	f'	message	r'
IDLE	IDLE	C		X	IDLE	MOVE	filament change in progress	3
IN_PROG	IDLE	C		X	PAUSED	MOVE	filament change in progress	3
X	MOVE		ok_MOVE	X	X	WIPE	moving to wiping location successful	3
X	MOVE		fail_MOVE	3	X	MOVE	failure, trying again	2
X	MOVE		fail_MOVE	0	X	IDLE	process failed, initiate again	3
PAUSED	HEAT		ok_HEAT	X	IN_PROG	IDLE	nozzle heating successful	3
X	WIPE	C	X	X	X	WIPE	filament change already in progress	X

The model was checked against a set of different test cases (`test_cases.csv`).

2.1.2 Transition

The two sub-machines are shown in Figures 9 and 10. Transitions are labelled as *input* [*guard condition*] / *action*, where r is the number of retries left. A C or F event starts the replacement only when the filament replacement machine is in its idle state; while a replacement is running, further C or F events are ignored (self-loop with the message "filament change already in progress"). If a C or F event arrives in the same step as a feedback signal, the external event takes precedence and the feedback is discarded.

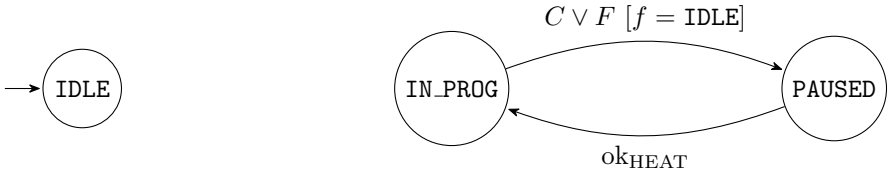


Figure 9: Printing sub-machine with state $p \in P$. The transition from `IDLE` to `IN_PROG` (print started) belongs to the print job handling and is not modelled here.

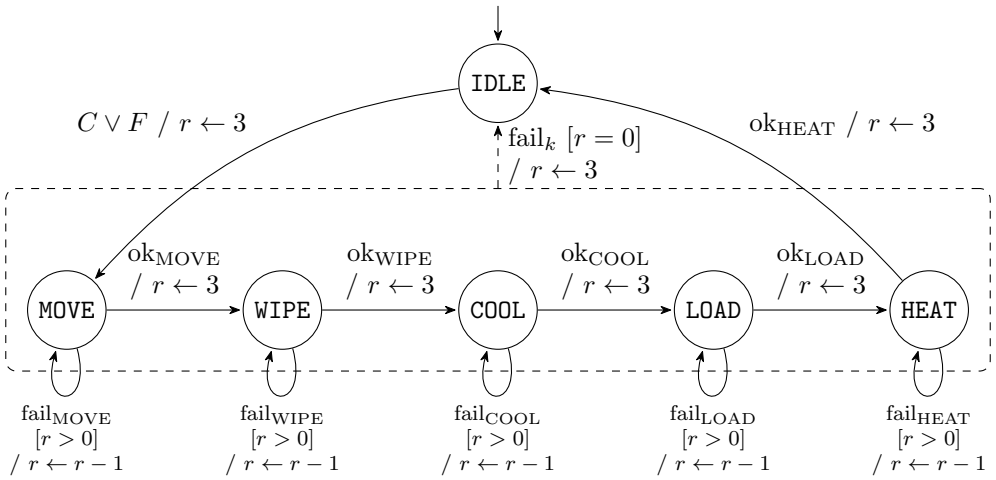


Figure 10: Filament replacement sub-machine with state $f \in \mathcal{F}$. For readability the five fail_k [$r = 0$] transitions from the steps k back to `IDLE` are drawn as a single dashed arrow from the box around the steps. The two sub-machines are synchronised: the $C \vee F$ event and the ok_{HEAT} event act on both machines in the same step.

2.2 Parked Detection

The head is parked at the origin when both the horizontal and the vertical motion have hit their end of line, i.e. a motion was commanded but the corresponding motor did not rotate. The detector is the Deterministic Finite Automaton (DFA)

$$\mathcal{A} = (Q_A, \Sigma_A, \delta_A, q_{A,0}, F_A) \tag{2.4}$$

with

$$\begin{aligned} Q_A &= \{q_0, q_1, q_2, q_3\}, \\ \Sigma_A &= M \times m = \{(H,+), (H,0), (V,+), (V,0)\}, \\ q_{A,0} &= q_0, \qquad F_A = \{q_3\}, \end{aligned} \tag{2.5}$$

where the states have the meaning

- q_0: no end of line detected,
- q_1: horizontal end of line (left side) detected,
- q_2: vertical end of line (bottom side) detected,
- q_3: both detected, the head is parked.

The transition function $\delta_A : Q_A \times \Sigma_A \to Q_A$ is given in Table 3 and drawn in Figure 11.

Table 3: Transition function δ_A of the parked detector.

q	$(H,+)$	$(H,0)$	$(V,+)$	$(V,0)$
$\to q_0$	q_0	q_1	q_0	q_2
q_1	q_0	q_1	q_1	q_3
q_2	q_2	q_3	q_0	q_2
$* q_3$	q_2	q_3	q_1	q_3

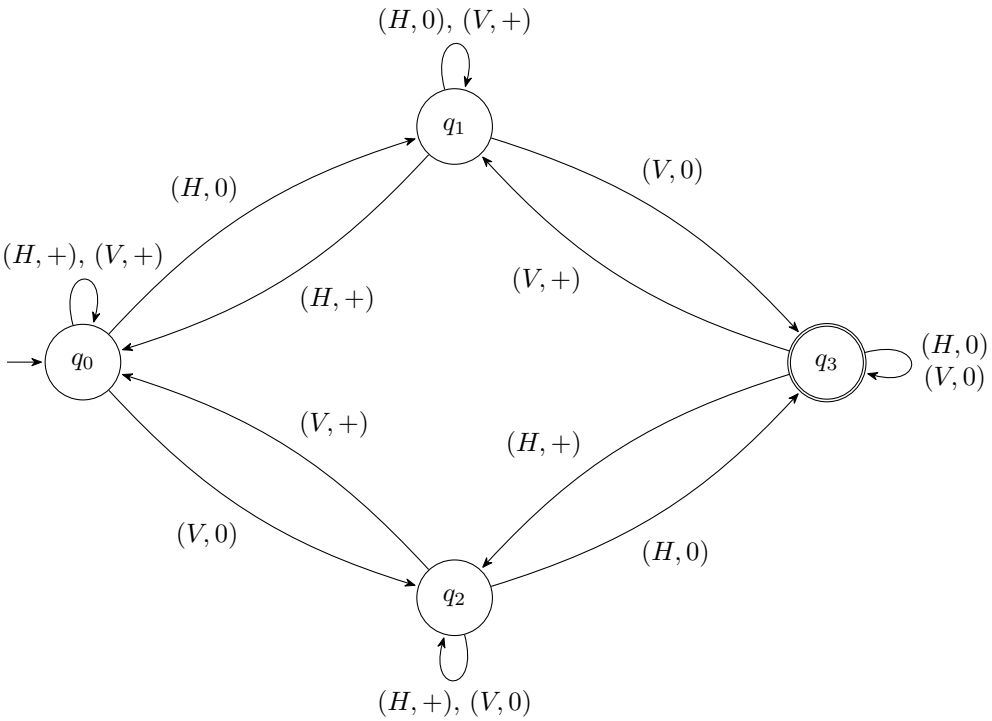


Figure 11: Transition of the parked detector $\mathcal{A}$. The accepted language is the set of sequences σ that end in q_3.

2.3 Detecting the Other Corners

The automaton is extended to the motions H_x, H_y and V and to the angular velocities $+$, $-$ and 0,

$$\begin{aligned} \mathcal{A}' &= (Q_A, \Sigma'_A, \delta'_A, q_0, F_A), \\ \Sigma'_A &= \{H_x, H_y, V\} \times \{+, -, 0\}, \end{aligned} \tag{2.6}$$

with Q_A, q_0 and F_A as before. The transition function $\delta'_A : Q_A \times \Sigma'_A \to Q_A$ is given in Table 4 and drawn in Figure 12.

Table 4: Transition function δ'_A of the corner detector.

q	$(H_x, \pm)$	$(H_x, 0)$	$(H_y, \pm)$	$(H_y, 0)$	$(V, \cdot)$
$\to q_0$	q_0	q_1	q_0	q_2	q_0
q_1	q_0	q_1	q_1	q_3	q_1
q_2	q_2	q_3	q_0	q_2	q_2
$*\, q_3$	q_2	q_3	q_1	q_3	q_3

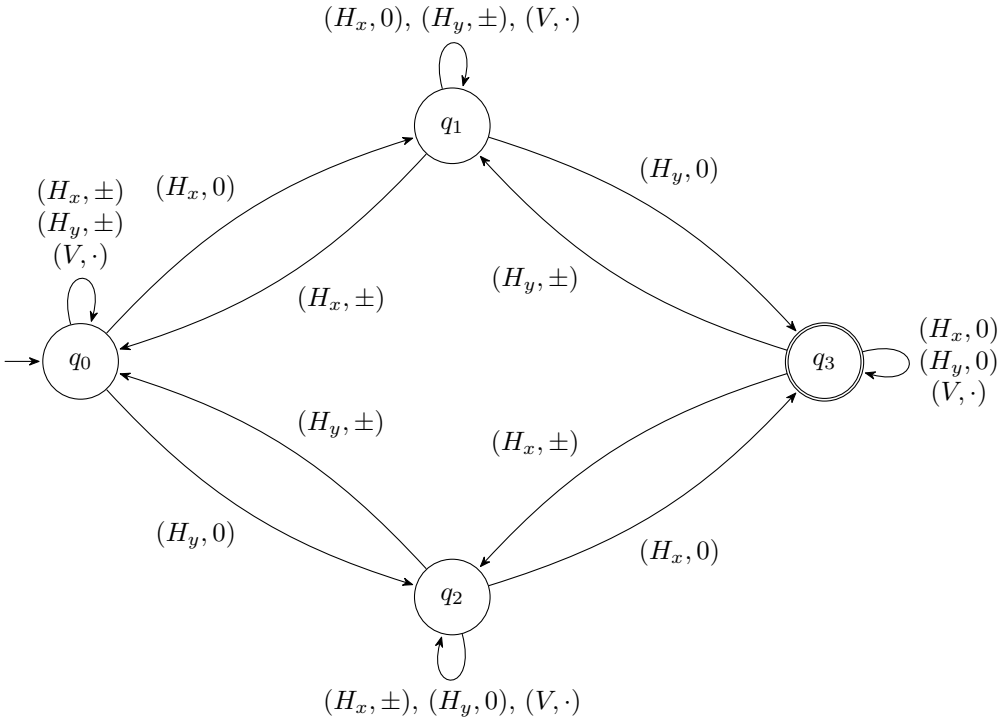


Figure 12: Transition of the corner detector $\mathcal{A}'$.

3 Hybrid System

3.1 Q3.1: Filament Replacement

Hybrid model: state, inputs, modes, events, invariants, guards and reset maps (tables).

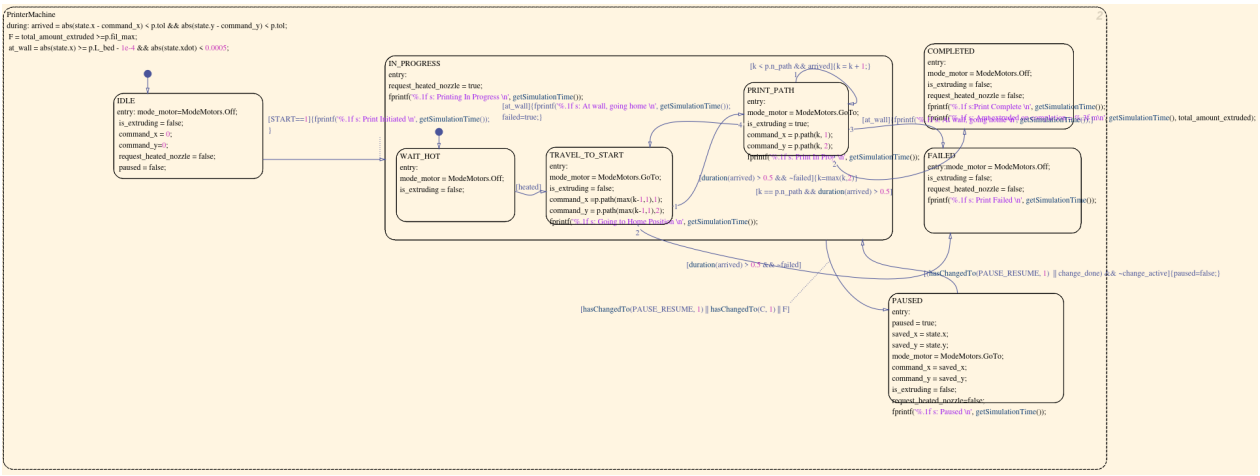


Figure 13: Stateflow chart of the printer machine.

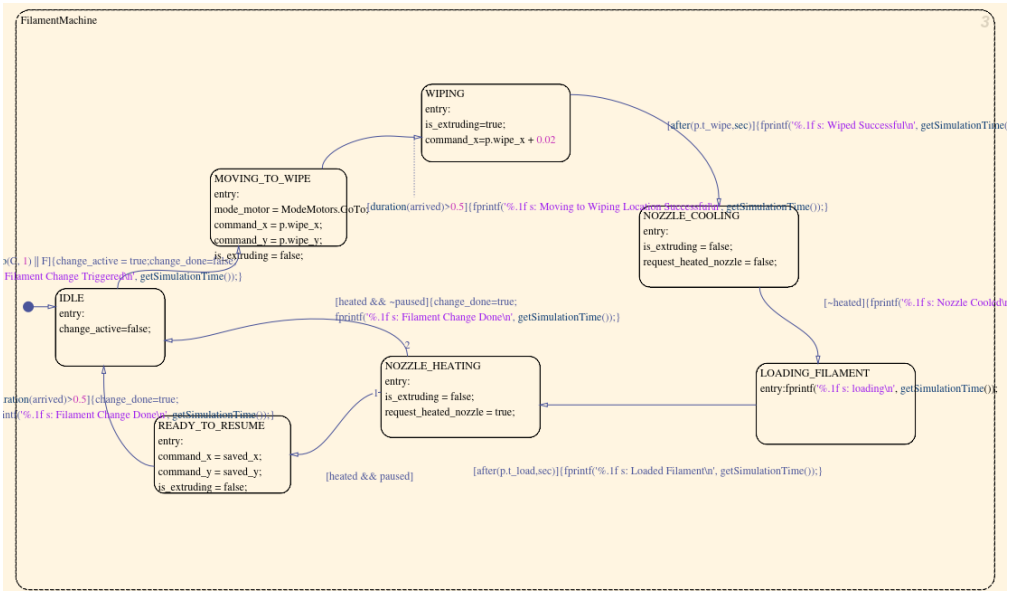


Figure 14: Stateflow chart of the filament machine.

Simulation setup: square path, L_{spool}, the instants at which C and F are applied.

3.1.1 Case 1: Colour Change

Scenario: C is applied at $t \approx 69.6\,\text{s}$ while printing; the head moves to the wipe location, wipes, loads the new filament, reheats and resumes (red to green). The run completes at $t \approx 163\,\text{s}$.

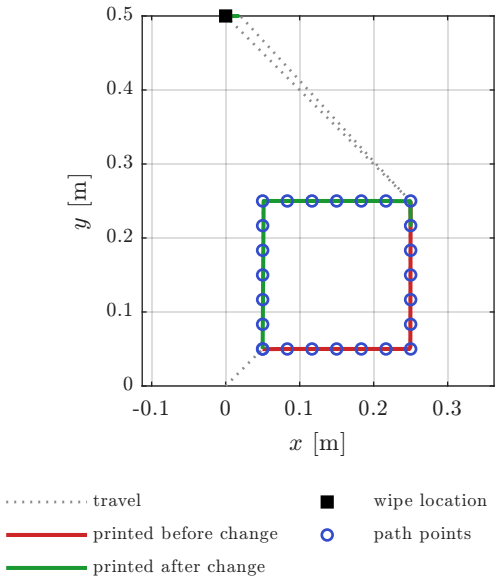
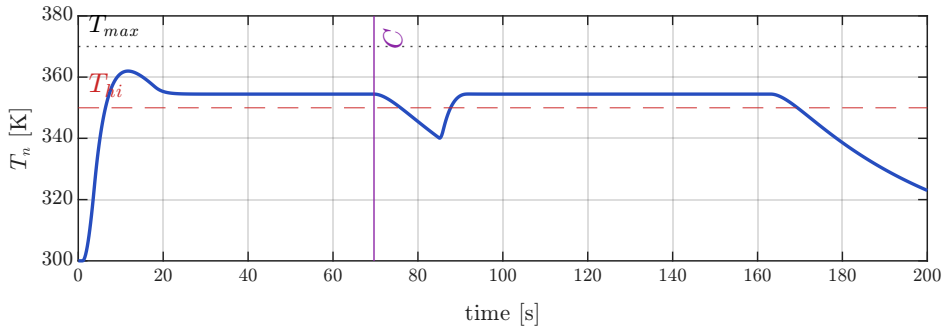
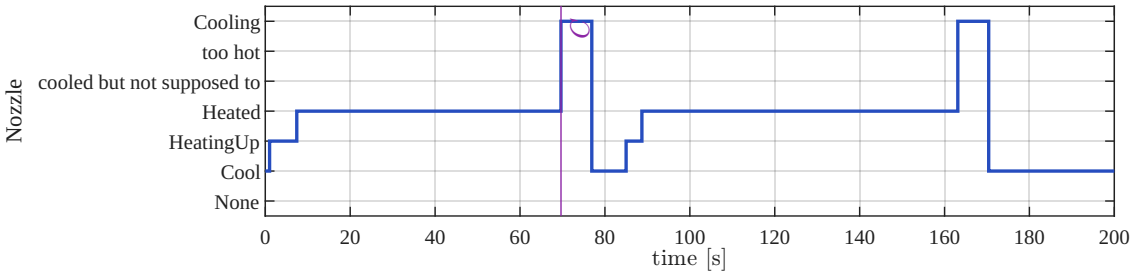


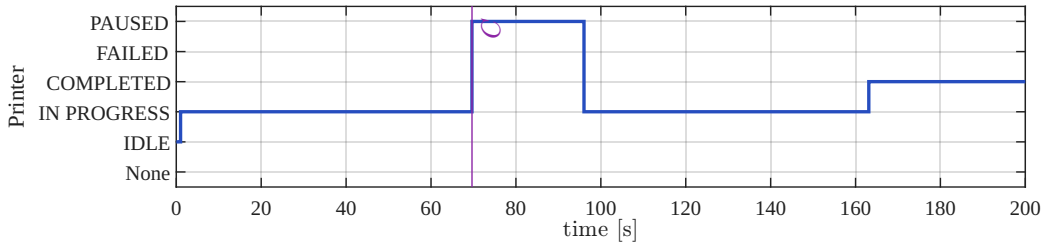
Figure 15: Case 1. Head path: printed before the change (red), after (green), and travel moves.



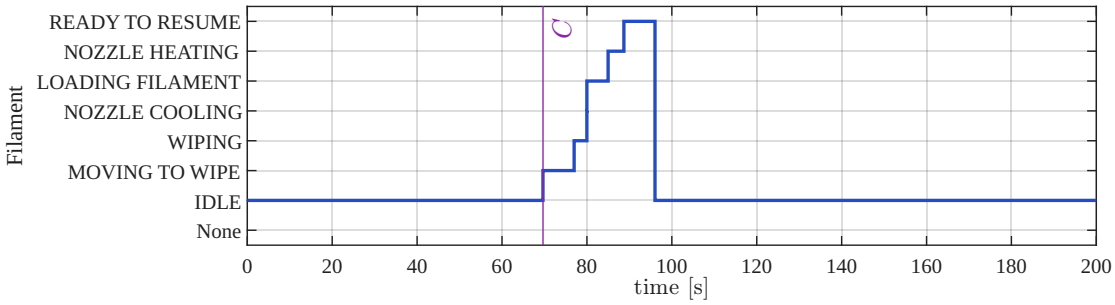
(a) Nozzle temperature T_n with the thresholds T_{hi} and T_{max}.



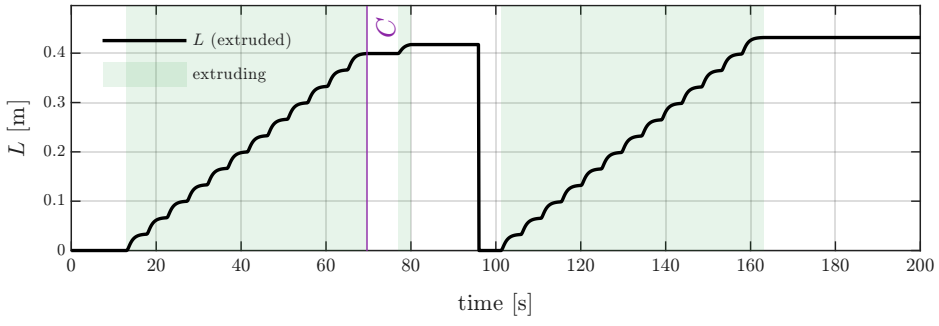
(b) Nozzle machine.



(c) Printer machine.



(d) Filament machine.



(e) Extruded filament length L against L_{spool}.

3.1.2 Case 2: Filament Runout

Scenario: the spool runs out (F) at $t \approx 69.6\,\text{s}$, and again at $t \approx 158\,\text{s}$ after the first replacement, so the filament sequence runs twice. The run completes at $t \approx 191\,\text{s}$.

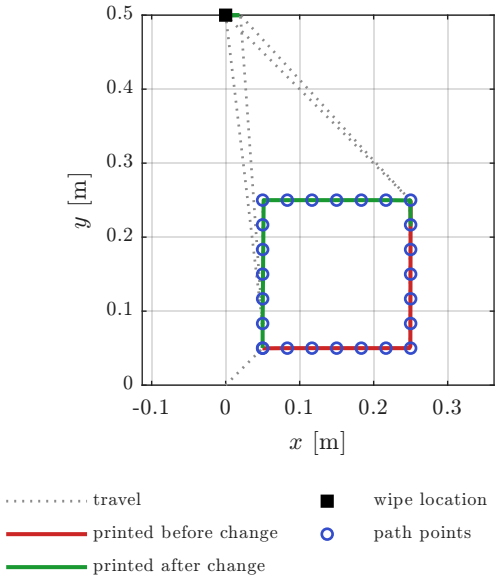
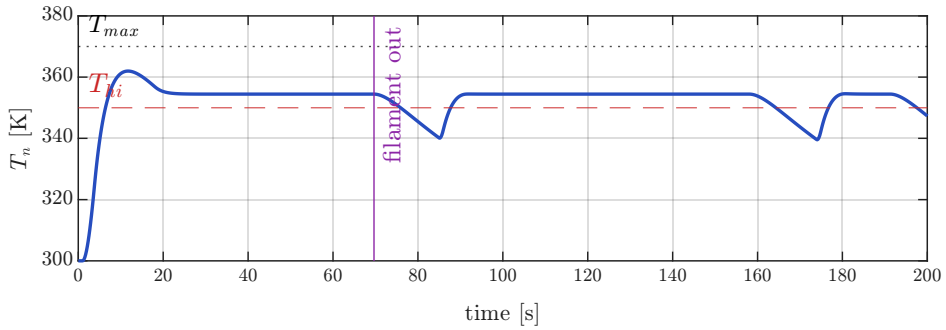
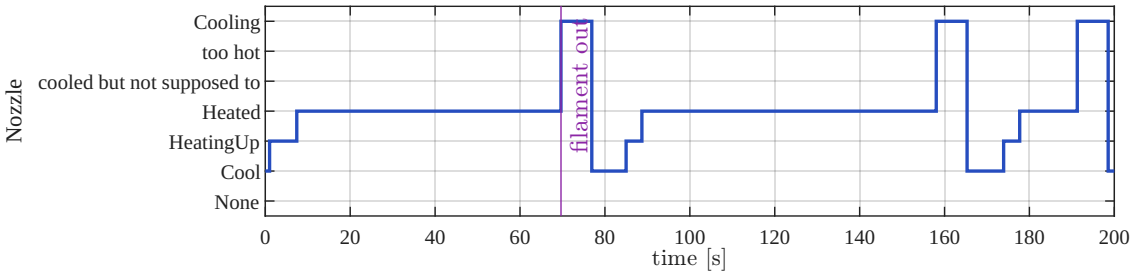


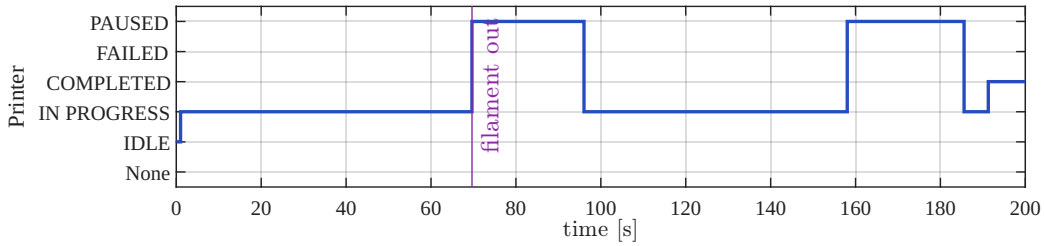
Figure 17: Case 2. Head path: printed before the change (red), after (green), and travel moves.



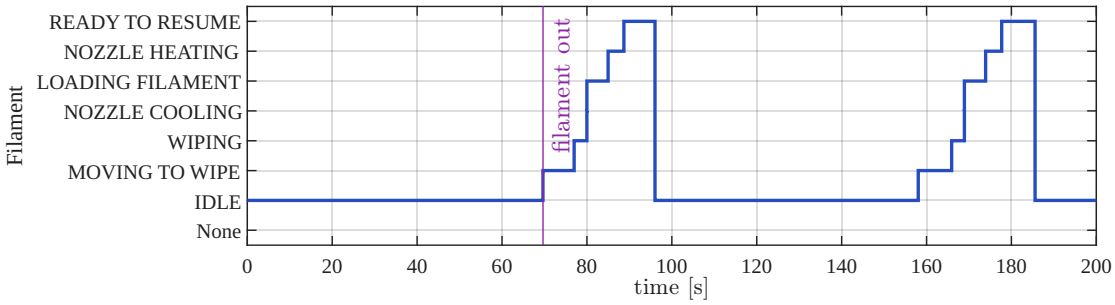
(a) Nozzle temperature T_n with the thresholds T_{hi} and T_{max}.



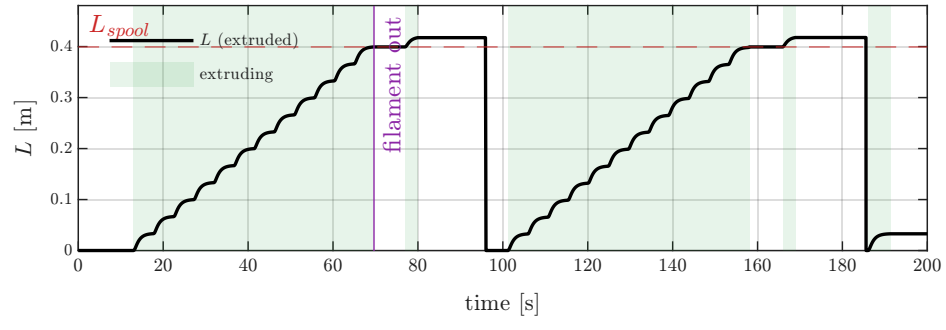
(b) Nozzle machine.



(c) Printer machine.



(d) Filament machine.



(e) Extruded filament length L against L_{spool}.

3.1.3 Discussion

Compare both cases with the FSM of Chapter 2: order of filament states, nozzle cooling and reheating, resumption point.

3.2 Q3.2: Drawing a Square

Draw the square using only the motor angle sensors; add the rotated square (bonus).

References